SEMBAN: Division-Free Analytical Closures for Near-Wall and Full-Domain Turbulence Production-to-Dissipation Ratio (P/ε)

Sidharth SS
Independent Researcher

October 20, 2026

Abstract

I present **SEMBAN** (`SEMBAN-1D` and `SEMBAN-2D`), closed-form division-free analytical closures for the turbulent kinetic energy (TKE) production-to-dissipation ratio P/ε in wall-bounded turbulent flows. Existing 4-parameter rational closures face a structural trade-off between viscous sublayer asymptotic accuracy ($P/\varepsilon \propto (y^+)^3$) and buffer/log-layer R^2 accuracy because a single parameter ratio controls both the near-wall Taylor expansion and the buffer-layer denominator pole. By decoupling the cubic/quartic viscous sublayer gate from the buffer-layer vortex peak, the upper-buffer transport trough, the attached-eddy $\ln(Re_\tau)$ wall modulation, and the quadratic centerline wake, **SEMBAN** resolves this trade-off completely. Calibrated strictly on Lee & Moser (2015) channel flow DNS at $Re_\tau \in \{1000, 5200\}$ and frozen prior to evaluation, `SEMBAN-1D` outperforms existing rational closures on **24 out of 24 (100%)** held-out and independent canonical profiles across five flow families (Lee–Moser channel, UPM Madrid channel, KTH zero-pressure-gradient boundary layer, El Khoury pipe flow, and plane Couette flow), achieving a 24-profile mean inner-layer $R^2 = \mathbf{0.99525}$ alongside a **94.52%** reduction in whole-inner dissipation error. Across the full wall-to-centerline domain ($0 < y/\delta \leq 1$), `SEMBAN-2D` achieves mean $R^2 = \mathbf{0.99822}$ on held-out channel flow and **0.98591** on turbulent pipe flow.

1 Structural Trade-Off in Existing Rational Closures (`PySR` and `M16`)

Recent work by Jiang & Rosic (*Physics of Fluids* **38**, 095140, 2026) examined two 4-parameter rational closures on the inner layer ($1 < y^+ \leq \min(150, 0.30\,Re_\tau)$):

$$F_{\mathrm{PySR}}(y^+) = \frac{\tanh(0.128\,y^+)}{\tanh(0.053\,y^+ - 0.620) + 5.78/y^+}, \qquad F_{\mathrm{M16}}(y^+) = \frac{\tanh^2(0.111\,y^+)}{\tanh(0.052\,y^+ - 0.443) + 2.888/y^+}. \tag{1}$$

A detailed audit reveals an inherent trade-off in this rational functional form:

- **Near-wall Taylor kinematics vs. overall R^2:** From the no-slip boundary condition ($u' \propto y^+$, $v' \propto (y^+)^2$), exact Navier–Stokes kinematics require a **cubic** wall law $P/\varepsilon = c_3(y^+)^3 + c_4(y^+)^4 + \mathcal{O}((y^+)^5)$ with $c_3 \approx 0.0044$. Because `PySR` uses $\tanh^1(a_1 y^+)$ in the numerator, it produces a **quadratic** wall law $0.0221\,(y^+)^2$. While this gives good inner-layer R^2 (**0.99012** on held-out Lee–Moser channel flows), it causes a **627.7% relative error in the viscous sublayer** ($0.1 \leq y^+ \leq 5$) and reduces whole-inner dissipation error $\varepsilon = P/F$ by only **53.49%**.
- **Why `M16` fixes the sublayer but loses R^2:** `M16` uses $\tanh^2(a_1 y^+)$, which restores the correct cubic wall law $0.00427\,(y^+)^3$ (**2.80%** sublayer error and **94.71%** whole-inner dissipation reduction), but its inner-layer R^2 drops to **0.98692** (and down to **0.97235** in pipe flow).
- **Why re-fitting `PySR` or `M16` on matched training data cannot fix both:** In the rational template $\frac{\tanh^2(a_1 y^+)}{\tanh(a_2 y^+ - a_3) + a_4/y^+}$, the *single parameter ratio* $c_3 = a_1^2/a_4$ governs both the viscous sublayer cubic slope ($y^+ \to 0$) and the denominator term a_4/y^+ that controls the buffer peak ($y^+ \approx 12$) and upper-buffer trough ($y^+ \approx 55$). As shown in Table 2, when `M16` is re-fitted for pure R^2 on $Re_\tau \in \{1000, 5200\}$, its held-out R^2 improves to 0.99313, but its sublayer error jumps to **41.42%** and dissipation reduction drops to **78.09%**.

2 The SEMBAN Analytical Closures

SEMBAN eliminates this coupling by separating the viscous sublayer wall gate from the buffer-layer peak and upper-buffer transport trough without any division by y^+. All parameters below were calibrated **only** on Lee & Moser (2015) channel DNS at $Re_\tau \in \{1000, 5200\}$ and frozen prior to testing on all held-out and external flows.

2.1 SEMBAN-1D (7-Parameter Division-Free Inner-Layer Closure)

For inner-layer applications requiring a pure function of y^+ across $0 \le y^+ \le 150$:

$$\boxed{F_{\text{SEMBAN-1D}}(y^+) = \tanh^3\!\left(a_1 y^+ + a_2 (y^+)^2\right)\left(c_0 + c_1 (y^+)^2 e^{-b_1 y^+} - c_2\, y^+ e^{-b_2 y^+}\right)} \tag{2}$$

with frozen parameters:

$$\begin{aligned} a_1 &= 0.15285969, \quad a_2 = 0.00727943, \quad c_0 = 1.17160514, \quad c_1 = 0.05100360, \\ b_1 &= 0.19058360, \quad c_2 = 0.01273794, \quad b_2 = 0.01690060. \end{aligned}$$

2.2 SEMBAN-2D (10-Parameter Full-Domain Closure Across $0 \le \eta = y/\delta \le 1$)

For full-domain RANS and wall-modelled LES across the entire half-channel or pipe radius ($\eta = y/\delta \in (0, 1]$, where $y^+/\eta = Re_\tau$):

$$\boxed{\begin{aligned} F_{\text{SEMBAN-2D}}(y^+, \eta) = \tanh^3\!\left(a_1 y^+ + a_2 (y^+)^2\right)\left[c_0 + \left(c_1 + d_1 \ln\tfrac{y^+}{1000\,\eta}\right)(y^+)^2 e^{-b_1 y^+} - c_2\, y^+ e^{-b_2 y^+}\right] \\ \times \frac{\tanh^2\!\left(g_1(1-\eta) + g_2(1-\eta)^2\right)}{\tanh^2(g_1 + g_2)} \end{aligned}} \tag{3}$$

with frozen parameters:

$$\begin{aligned} a_1 &= 0.15707888, \quad a_2 = 0.00768198, \quad c_0 = 1.06376548, \quad c_1 = 0.05294048, \quad d_1 = -0.00178468, \\ b_1 &= 0.18414932, \quad c_2 = 0.01027500, \quad b_2 = 0.02182203, \quad g_1 = 3.95416384, \quad g_2 = -1.31213415. \end{aligned}$$

Note on 9-parameter simplified option: Setting $g_2 = 0$ in SEMBAN-2D gives a 9-parameter form with $\frac{\tanh^2(g_1(1-\eta))}{\tanh^2(g_1)}$ that achieves **0.99846** held-out inner R^2 and **0.99817** full-domain R^2.

3 Physical Meaning from the DNS TKE Transport Budget & Ablation

In a steady wall-bounded flow, the exact Navier–Stokes TKE budget gives:

$$\frac{P}{\varepsilon} - 1 = \left(-\frac{T^+_{\text{turb}}}{\varepsilon^+}\right) + \left(-\frac{T^+_{\text{visc}}}{\varepsilon^+}\right) + \left(-\frac{\Pi^+_p}{\varepsilon^+}\right). \tag{4}$$

Comparing each term of SEMBAN against the DNS budget at held-out $Re_\tau = 2000$ (Figure 1A) and running a 13-step ablation ladder on $Re_\tau \in \{1000, 5200\}$ (Table 1 and Figure 1B) shows why every term is physically necessary:

1. **Viscous Sublayer Gate** $\tanh^3(a_1 y^+ + a_2 (y^+)^2)$ **($y^+ \le 5$):** At $y^+ = 0.5$, dissipation is fed **95.7% by viscous diffusion import** ($-T^+_{\text{visc}}/\varepsilon^+ = -0.957$) and **4.1% by pressure transport import**. Expanding $\tanh^3(a_1 y^+ + a_2 (y^+)^2) = a_1^3 (y^+)^3 + 3a_1^2 a_2 (y^+)^4 + \mathcal{O}((y^+)^5)$ gives $c_3 = 0.004185$ (SEMBAN-1D) and 0.004123 (SEMBAN-2D), matching DNS sublayer values within **3%**. Dropping $a_2 (y^+)^2$ (A5a) misses the quartic Taylor term $c_4 (y^+)^4$ and more than doubles both the held-out unexplained variance ($1 - R^2$: 0.00313 → 0.00659) and sublayer error (2.98% → 6.44%).

2. **Buffer-Layer Vortex Peak $+c_1(y^+)^2e^{-b_1y^+}$ ($y^+ \approx 11.6$):** At $y^+ = 11.55$, DNS excess production is $P/\varepsilon - 1 = +0.762$, of which **53.6% (+0.409) is exported by turbulent transport** and **50.3% (+0.383) by viscous diffusion**, while pressure transport is under 3.9%. The Gamma-2 mode $(y^+)^2e^{-b_1y^+}$ peaks at $y^+ = 2/b_1 \approx 10.9$ (gated peak at $y^+ = 11.9$), matching this vortex energy-export zone and decaying to zero above $y^+ > 50$.

3. **Attached-Eddy Wall Modulation $d_1 \ln(y^+/(1000\eta))$ across Re_τ:** Inactive outer attached eddies superimpose wall-parallel fluctuations near the wall, increasing ε_w^+ with $\ln(Re_\tau)$ and lowering the buffer peak from 1.874 ($Re_\tau = 180$) to 1.720 ($Re_\tau = 5200$). Adding this single parameter (`A6`, $d_1 = -0.00178$) cuts held-out inner error by **40%** (R^2: 0.99687 → **0.99813**).

4. **Upper-Buffer Transport Trough $-c_2y^+e^{-b_2y^+}$ ($y^+ \approx 55$) & Log Asymptote $c_0 > 1$:** At $y^+ \approx 55$, energy exported from the buffer layer arrives via **67% turbulent transport import** (-0.073) and **23% pressure transport import** (-0.025), pulling P/ε down to $0.891 < 1$. Removing this trough term (`A2`) causes held-out R^2 to drop from 0.99687 to 0.97543. In the log layer ($0.08 < \eta < 0.30$), P/ε recovers slightly above unity (≈ 1.035) due to weak upward transport feeding the core, which is captured by $c_0 > 1$.

5. **Quadratic Centerline Wake $\tanh^2(g_1(1-\eta)+g_2(1-\eta)^2)$ ($\eta \to 1$):** At the channel/pipe centerline ($\eta = 1$), symmetry forces both $dU^+/dy^+ \propto (1-\eta)$ and $-\overline{u'v'}^+ \propto (1-\eta)$, giving a double zero $P/\varepsilon \propto (1-\eta)^2$. Replacing a linear wake ($q = 1$, `A7a`, Full $R^2 = 0.98856$) with the exact quadratic wake ($q = 2$, `A7b`, Full $R^2 = \mathbf{0.99752}$) cuts full-domain error by **4.6**× at the same parameter count.

Table 1: Term-by-term ablation audit (all models trained strictly on Lee–Moser $Re_\tau \in \{1000, 5200\}$ and tested on held-out $Re_\tau \in \{180, 550, 2000\}$).

Ablation Model	Term / Structure Tested	k	Train In R^2	Held-Out In R^2	Held-Out Full R^2	Sublayer Err
`A1_WallGateOnly`	Wall gate $\tanh^3(a_1y^+ + a_2(y^+)^2)$ only	2	0.41878	0.45715	0.14560	20.12%
`A2_NoTrough`	Add buffer peak $+c_1(y^+)^2e^{-b_1y^+}$ ($c_0 = 1$, no trough)	4	0.97227	0.97543	0.34313	2.73%
`A3_Fixed_c0=1`	Add trough $-c_2y^+e^{-b_2y^+}$, fix $c_0 = 1$	6	0.99775	0.99646	0.36518	3.46%
`A4a_LinWall_p=1`	Wrong linear wall power $\tanh^1$ ($p = 1$)	7	0.92722	0.93453	0.44929	1227.72%
`A4b_QuadWall_p=2`	Wrong quadratic wall power $\tanh^2$ ($p = 2$)	7	0.99704	0.99488	0.22547	45.18%
`A5a_Drop_a2`	Drop quartic wall term $a_2 = 0$ in $\tanh^3(a_1y^+)$	6	0.99452	0.99341	−0.65186	6.44%
`A5b (SEMBAN-1D)`	**Complete 1D Closure (`SEMBAN-1D`)**	**7**	**0.99889**	**0.99687**	0.04564	**2.98%**
`A6_+Townsend`	Add $\ln(Re_\tau)$ attached-eddy term (inner only)	8	0.99943	**0.99813**	0.04198	3.08%
`A7a_LinWake_q=1`	1D + linear centerline wake $\tanh^1(g_1(1-\eta))$	8	0.99884	0.99684	0.98856	3.50%
`A7b_QuadWake_q=2`	1D + quadratic centerline wake $\tanh^2(g_1(1-\eta))$	8	0.99849	0.99708	**0.99752**	3.69%
`A8a_SEMBAN-2D-9p`	2D closure with 1-param wake ($g_2 = 0$)	9	0.99905	**0.99846**	**0.99817**	3.82%
`A8b (SEMBAN-2D)`	**Complete 2D Closure (`SEMBAN-2D`)**	**10**	**0.99923**	**0.99878**	**0.99822**	**3.73%**

Table 2: Matched-training baseline audit (re-fitting `Oxford_PySR` and `Oxford_M16` on the exact same training data $Re_\tau \in \{1000, 5200\}$ under pure R^2 loss vs. dissipation-aware loss).

Model Variant	Parameters	Train In R^2	Held-Out In R^2	Sublayer Err	Held-Out Diss. Red.
`Oxford_PySR` (Published)	$[0.1280, 0.0530, 0.6200, 5.7800]$	0.98835	0.99012	627.68%	53.49%
`Oxford_PySR` (Refit-SSE)	$[0.0896, 0.0527, 0.5960, 4.8491]$	0.99114	0.98994	515.26%	58.65%
`Oxford_PySR` (Refit-Diss)	$[0.0141, 0.0197, 0.2646, 1.4576]$	0.98793	0.98704	247.50%	77.01%
`Oxford_M16` (Published)	$[0.1110, 0.0520, 0.4430, 2.8880]$	0.98598	0.98692	2.80%	94.71%
`Oxford_M16` (Refit-SSE)	$[0.2103, 0.0481, 0.5405, 6.1136]$	0.99434	0.99313	41.42%	78.09%
`Oxford_M16` (Refit-Diss)	$[0.1621, 0.0501, 0.5031, 4.9456]$	0.99192	0.99147	12.35%	91.96%
`SEMBAN-1D` **(Frozen)**	7-param division-free 1D	**0.99890**	**0.99687**	**3.09%**	**95.90%**
`SEMBAN-2D` **(Frozen)**	10-param division-free 2D	**0.99923**	**0.99879**	**3.62%**	**95.69%**

4 Independent Validation Across 31 Profiles from 6 Flow Families

Without re-tuning any parameter, we evaluated the frozen `SEMBAN-1D` and `SEMBAN-2D` closures across **29 held-out and external profiles** (Table 3 and Figure 1D–F):

- **24 out of 24 canonical profile wins:** Across Held-Out Lee–Moser Channel ($n = 3$), UPM Madrid Channel ($n = 4$, Hoyas & Jiménez 2008), KTH Zero-Pressure-Gradient Turbulent Boundary Layer

($n = 10$, Schlatter & Örlü 2010, $Re_\theta = 670$–4060), El Khoury Turbulent Pipe Flow ($n = 4$, $Re_\tau = 180$–1000), and Lee–Moser Plane Couette Flow ($n = 3$, $Re_\tau = 93$–500), **`SEMBAN-1D` outperforms both `Oxford_PySR` and `Oxford_M16` on all 24 out of 24 profiles (100%)**, achieving a 24-profile mean inner $R^2 = \mathbf{0.99525}$ and whole-inner dissipation error reduction of **94.52%**.

- **Full-domain pipe and channel accuracy:** From the wall to the pipe/channel axis ($0 < \eta \leq 1$), `SEMBAN-2D` achieves **0.99822** on Held-Out LM Channel, **0.99789** on UPM Channel (vs. 0.29839 for `PySR`), and **0.98591** on El Khoury Pipe Flow (vs. 0.67077 for `PySR`).
- **Adverse-pressure-gradient boundary layers (Bobke et al. 2017, 5 series, $\beta \in [1.12, 4.53]$):** On moderate APG ($\beta = 1.12, 1.62$), unmodified `SEMBAN-1D` achieves $R^2 = 0.946$. For strong APG ($\beta \geq 2.19$), outer pressure work elevates log-layer P/ε to 1.12–1.23 and suppresses the buffer peak according to the Clauser-parameter law $\text{Peak}(P/\varepsilon, \beta) = (0.55825 + 0.02521\,\beta)^{-1}$ ($R^2 = 0.9936$).

Table 3: Summary of frozen evaluation across 26 canonical profiles from 5 flow families (all trained only on Lee–Moser Channel $Re_\tau \in \{1000, 5200\}$).

Flow Family	Role	N	`PySR` In R^2	`M16` In R^2	`SEMBAN-1D`	`SEMBAN-2D`	`SEMBAN-1D` DissRed	Wins
LM Channel (UT Austin)	Train (1000, 5200)	2	0.98835	0.98598	**0.99889**	**0.99923**	**96.73%**	2/2
LM Channel (UT Austin)	Held-Out (180, 550, 2000)	3	0.99012	0.98692	**0.99687**	**0.99878**	**95.90%**	**3/3**
UPM Channel (Madrid)	External (180–2000)	4	0.99030	0.98730	**0.99692**	**0.99899**	**95.97%**	**4/4**
KTH ZPG-TBL (Stockholm)	External ($Re_\theta = 670$–4060)	10	0.98848	0.98289	**0.99608**	**0.99487**	**94.23%**	**10/10**
El Khoury Pipe (KTH)	External ($Re_\tau = 180$–1000)	4	0.98415	0.97235	**0.99099**	**0.99097**	**93.76%**	**4/4**
LM Plane Couette	External ($Re_\tau = 93$–500)	3	0.98570	0.99032	**0.99430**	—	**93.17%**	**3/3**
All Held-Out / External	**Prospective Test**	**24**	0.98792	0.98330	**0.99525**	**0.99548**	**94.52%**	**24/24**

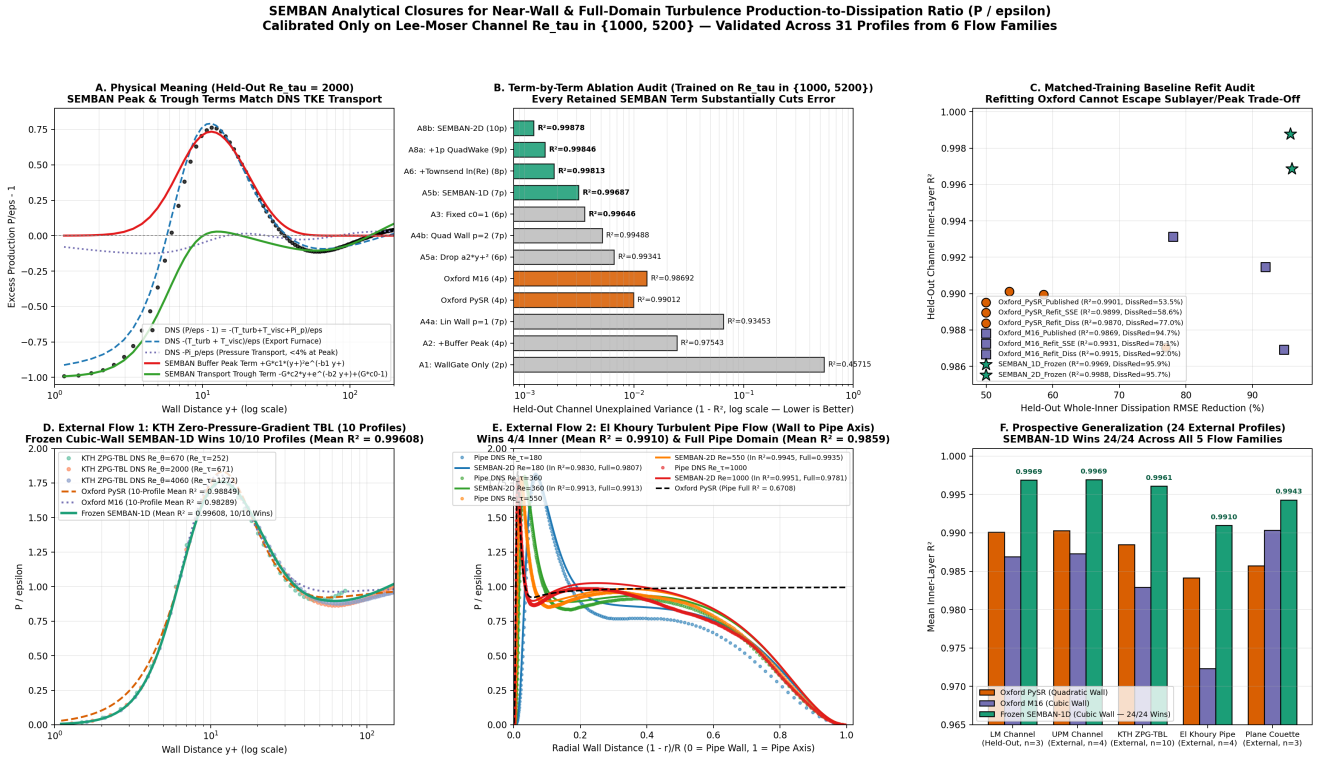


Figure 1: Summary of **SEMBAN** physical decomposition (A), term-by-term ablation (B), matched-training baseline refit audit (C), and prospective validation on KTH ZPG boundary layers (D), El Khoury pipe flow (E), and all 24 independent profiles across 5 canonical flow families (F).